

All Metaplectic Compact-Symbol Classes Are Identifiable

Removing the diagonalizability hypothesis in arXiv:2411.01840

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Status: full solution likely valid. The proof removes the simultaneous-diagonalizability hypothesis from Theorem 5.11 of McNulty's source paper. Human review is recommended for the metaplectic block conventions.

1 Source problem and result

McNulty studies identification of Hilbert–Schmidt operators from the diagonal of their Gabor matrix [1]. In the notation of that paper, for $\mathcal{A} \in Sp(2d, \mathbb{R})$ and compact $K \subset \mathbb{R}^{2d}$, set

$$\mathcal{P}_{\mathcal{A}, K} = \{T \in \mathcal{HS} : \text{supp}(\sigma_T^{\mathcal{A}}) \subset K\}, \quad \sigma_T^{\mathcal{A}} = \mu(\mathcal{A})\sigma_T.$$

Here σ_T is the Weyl symbol and $\mu(\mathcal{A})$ is the metaplectic operator acting on symbols. The class is *identifiable* on a lattice Λ with window $S_0 = \varphi_0 \otimes \varphi_0$ when

$$T \longmapsto \{Q_{S_0}T(\lambda, \lambda) : \lambda \in \Lambda\}$$

is injective on $\mathcal{P}_{\mathcal{A}, K}$.

The source theorem assumes simultaneous diagonalizability of two matrices in a symplectic decomposition:

Theorem 5.11. *Let $\mathcal{A} \in Sp(2d, \mathbb{R})$, such that $A, \tilde{B}$ corresponding to $\mathcal{A}$ in the decomposition in Lemma 5.10 are simultaneously diagonalisable. Then for any compact $K \subset \mathbb{R}^{2d}$, there exists some lattice $\Lambda = A\mathbb{Z}^{2d}$ such that $\mathcal{P}_{\mathcal{A}, K}$ is identifiable on Λ with respect to the Gaussian $S_0 = \varphi_0 \otimes \varphi_0$.*

Proof. Since the metaplectic operators are unitary,

$$\begin{aligned} \langle T, \alpha_x(S) \rangle &= \langle \sigma_T, T_x \sigma_S \rangle \\ &= \langle \mu(\mathcal{A})\sigma_T, \mu(\mathcal{A})T_x \sigma_S \rangle. \end{aligned}$$

We decompose $\mathcal{A}$ as in Lemma 5.10:

$$\mathcal{A} = D_L V_{\tilde{A}} V_{\tilde{B}}^T V_A.$$

Inserting this into the inner product gives

$$\langle T, \alpha_x(S) \rangle = \langle \mu(D_L V_{\tilde{A}})^* \mu(\mathcal{A})\sigma_T, \mu(V_{\tilde{B}}^T V_A)T_x \sigma_S \rangle.$$

Since $\mu(D_L V_{\tilde{A}})^*$ maps compactly supported functions on K unitarily to compactly supported functions on $L^{-1} \cdot K$, if $\mu(V_{\tilde{B}}^T V_A)T_x \sigma_S$ is dense in $L^2(L^{-1} \cdot K)$ then we have operator identification. Since K is an arbitrary compact subset, we may consider K instead of $L^{-1} \cdot K$ here for simplicity of notation.

We consider the window S is given by

$$\sigma_S(t) = e^{-\pi t \cdot t}$$

The final remark asks whether the hypothesis can be relaxed:

Thus, it can easily be seen that the ability to identify, and consequently perform phase retrieval, applies to any of the sets of operators $S = f \otimes f$ where $\mu(A)f$ is compactly supported for any $A \in Sp(d, \mathbb{R})$.

Remark 5.14. The condition of simultaneously diagonalisable $A, \tilde{B}$ in Theorem 5.11 is required in order to make the real change of variables in Lemma 5.9. The complex analytical proof of the Müntz-Szász theorem relies on sets of uniqueness for complex analytic functions with a certain growth condition (cf. [27]). A general change of variables in Lemma 5.9 would not necessarily preserve this growth condition in the corresponding analytic function. It is possible that a similar result for sets of uniqueness for complex analytic functions of general exponential type would suffice to relax the diagonalisation condition in Theorem 5.11.

We remove it completely.

Theorem 1 (All-metaplectic identification). *Let $\mathcal{A} \in Sp(2d, \mathbb{R})$ be arbitrary and let $K \subset \mathbb{R}^{2d}$ be compact. There exists $\varepsilon > 0$ such that $\mathcal{P}_{\mathcal{A}, K}$ is identifiable on the real lattice*

$$\Lambda = \varepsilon \mathbb{Z}^{2d}$$

with respect to the standard Gaussian window $S_0 = \varphi_0 \otimes \varphi_0$. In particular, no simultaneous-diagonalizability hypothesis is needed in Theorem 5.11 of [1].

2 Proof intuition

The source proof explicitly computes a transformed Gaussian and then invokes a multivariable Müntz–Szász theorem. That computation creates the diagonalizability restriction. Two elementary observations avoid it.

First, any invertible complex matrix C produces a complete exponential lattice on a compact real set, provided the lattice spacing is small enough. Indeed, the basic exponentials $w_j(t) = e^{\varepsilon t^T C e_j}$ all lie in one logarithm chart. Laurent polynomials in the w_j 's therefore recover $\log w_j$, hence the linear forms $t^T C e_j$, hence every coordinate of t . Stone–Weierstrass then gives uniform density.

Second, a metaplectic image of a translated standard Gaussian is a fixed nonvanishing generalized Gaussian times exactly such a complex exponential. The coefficient matrix is $ZP + iR$, where $\operatorname{Re} Z > 0$ and $\begin{pmatrix} P \\ R \end{pmatrix}$ is the first block column of a symplectic matrix. Positivity and the Lagrangian relation force $ZP + iR$ to be invertible. The measurements in Theorem 1 are therefore inner products against a dense family on the compact support set.

3 The exponential-lattice lemma

Lemma 2 (Fine complex exponential lattices). *Let $m \geq 1$, let $C \in GL_m(\mathbb{C})$, and let $K \subset \mathbb{R}^m$ be compact. For all sufficiently small $\varepsilon > 0$,*

$$\operatorname{span}\{t \mapsto e^{t^T C \lambda} : \lambda \in \varepsilon \mathbb{Z}^m\}$$

is uniformly dense in $C(K)$. Consequently it is dense in $L^2(K)$.

Proof. Choose $\varepsilon > 0$ and $0 < r < 1$ so small that, for every $t \in K$ and $1 \leq j \leq m$,

$$\left| e^{\varepsilon t^T C e_j} - 1 \right| \leq r.$$

This is possible because K is compact and there are only finitely many columns of C . Put

$$w_j(t) = e^{\varepsilon t^T C e_j}.$$

Let $\mathcal{E}$ be the uniform closure in $C(K)$ of the span in the statement. Since multiplication adds lattice indices, $\mathcal{E}$ is a closed unital algebra, and it contains every Laurent monomial $w_1^{k_1} \cdots w_m^{k_m}$, $k \in \mathbb{Z}^m$.

The series

$$\log w_j = \sum_{q=1}^{\infty} \frac{(-1)^{q+1}}{q} (w_j - 1)^q$$

converges uniformly on K . Each partial sum is a polynomial in w_j , so $\log w_j \in \mathcal{E}$. By making ε smaller if necessary, $\varepsilon t^T C e_j$ lies in the logarithm chart about zero, and hence

$$\log w_j(t) = \varepsilon t^T C e_j.$$

Thus every component of the row vector $t^T C$ belongs to $\mathcal{E}$. Since C is invertible,

$$t^T = (t^T C) C^{-1},$$

so every real coordinate function $t \mapsto t_j$ belongs to $\mathcal{E}$.

It follows that $\mathcal{E}$ contains all complex polynomials in the real coordinate functions. Those polynomials form a self-adjoint unital algebra that separates the points of K ; the Stone–Weierstrass theorem therefore makes them uniformly dense in $C(K)$ [2]. Hence $\mathcal{E} = C(K)$. Finally, continuous functions are dense in $L^2(K)$, proving the last assertion. $\square$

4 Metaplectic Gaussian lattices

We use the source conventions for symmetric time–frequency shifts: $T_x = \rho(x, 0)$ and

$$\mu(\mathcal{A})\rho(z)\mu(\mathcal{A})^* = \rho(\mathcal{A}z).$$

The standard generators of the metaplectic group show that every centered standard Gaussian is taken to a generalized Gaussian

$$\mu(\mathcal{A})g_0(t) = c g_Z(t), \quad g_Z(t) = e^{-\pi t^T Z t},$$

where $c \neq 0$, $Z = Z^T \in M_m(\mathbb{C})$, and $\operatorname{Re} Z > 0$. For completeness: chirp multiplication adds a purely imaginary real-symmetric matrix to Z ; a real dilation sends Z to a real congruence; and the Fourier transform sends Z to Z^{-1} . All three operations preserve positive-definiteness of the real part, and these operators generate the metaplectic group [1].

Lemma 3 (Every metaplectic Gaussian lattice is complete locally). *Let $\mathcal{A} \in Sp(m, \mathbb{R})$ act metaplectically on $L^2(\mathbb{R}^m)$, and let $K \subset \mathbb{R}^m$ be compact. There is $\varepsilon > 0$ such that*

$$\operatorname{span}\{\mu(\mathcal{A})T_x g_0 : x \in \varepsilon \mathbb{Z}^m\}$$

is dense in $L^2(K)$.

Proof. Write $\mathcal{A}$ in block form

$$\mathcal{A} = \begin{pmatrix} P & Q \\ R & S \end{pmatrix}.$$

Metaplectic covariance gives

$$\mu(\mathcal{A})T_x g_0 = \rho(Px, Rx)\mu(\mathcal{A})g_0.$$

Expanding the translation and modulation of g_Z , and absorbing all nonzero factors depending only on x into c_x , yields

$$\mu(\mathcal{A})T_x g_0(t) = c_x g_Z(t) e^{2\pi i t^T (ZP + iR)x}. \quad (1)$$

We claim that $C := ZP + iR$ is invertible over $\mathbb{C}$. Suppose $Cu = 0$. Write $Z = X + iY$, where X, Y are real symmetric and $X > 0$. Multiplying $Cu = 0$ on the left by $(Pu)^*$ gives

$$(Pu)^* Z(Pu) + i(Pu)^* Ru = 0.$$

The symplectic relations imply that $P^T R$ is real symmetric. Therefore both $(Pu)^* Y(Pu)$ and $(Pu)^* Ru = u^* P^T Ru$ are real. Taking real parts in the preceding display gives

$$(Pu)^* X(Pu) = 0.$$

Since $X > 0$, $Pu = 0$. The equation $Cu = 0$ now gives $Ru = 0$. The first block column $\begin{pmatrix} P \\ R \end{pmatrix}$ of a symplectic matrix has full column rank, so $u = 0$. This proves $C \in GL_m(\mathbb{C})$.

Apply Lemma 2 to $2\pi C$ and K . Equation (1) differs from that dense exponential family only by nonzero scalars c_x and multiplication by g_Z . On compact K , the continuous function g_Z is bounded and bounded away from zero. Its multiplication operator is therefore invertible on $L^2(K)$, and the family in the statement is dense. $\square$

5 Proof of the identification theorem

Proof of Theorem 1. Put $m = 2d$, let $g_0 = \sigma_{S_0}$, and choose the lattice $\Lambda = \varepsilon \mathbb{Z}^m$ supplied by Lemma 3 for $\mathcal{A}$ and K . Suppose $T \in \mathcal{P}_{\mathcal{A}, K}$ has zero diagonal measurements on Λ . The Weyl-symbol identity used in the source proof gives, for every $\lambda \in \Lambda$,

$$\begin{aligned} 0 &= Q_{S_0} T(\lambda, \lambda) = \langle T, \alpha_\lambda(S_0) \rangle_{\mathcal{HS}} \\ &= \langle \sigma_T, T_\lambda g_0 \rangle_{L^2} = \langle \mu(\mathcal{A})\sigma_T, \mu(\mathcal{A})T_\lambda g_0 \rangle_{L^2}. \end{aligned}$$

By definition, $\mu(\mathcal{A})\sigma_T = \sigma_T^A$ is supported in K . Lemma 3 says that the second entries in the last inner products span a dense subspace of $L^2(K)$. Hence $\sigma_T^A = 0$. Metaplectic action and Weyl quantization are unitary on L^2 , so $T = 0$. Thus the measurement map is injective on $\mathcal{P}_{\mathcal{A}, K}$. $\square$

6 Verification, novelty, and limitations

The proof has no unproved dependency. Its convention-sensitive point is the sign in $ZP + iR$; reversing the modulation convention gives $ZP - iR$, and the same real-part argument proves invertibility. A numerical script in `code/sanity-check.py` tested 250 random symplectic frames and random Z with positive-definite real part; no near-singular matrix occurred. It also recovered coordinates from a truncated logarithm series to a maximum sample error of 2.86×10^{-14} . These checks are not part of the proof.

A bounded search of the run indexes, exact-title web/arXiv results, the source bibliography, and queries combining “metaplectic identification”, “compactly supported symbols”, “Gaussian lattice”, and “simultaneously diagonalisable” found no later removal of the source hypothesis. This is not an exhaustive priority claim.

The lattice depends on $\mathcal{A}$ and K . The theorem proves identifiability only; it does not supply frame bounds, stability, or a reconstruction series.

References

- [1] H. McNulty, *Metaplectic Quantum Time–Frequency Analysis, Operator Reconstruction and Identification*, arXiv:2411.01840, 2024.
- [2] W. Rudin, *Real and Complex Analysis*, 3rd ed., McGraw–Hill, 1987.